11. Based on the above passage, the following assumptions have been made:

1. For effective school education, parents have greater role than the governments.

2. School curriculum that conforms to today's requirements and is uniform for the entire country may address the issues brought out.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : Ans. (d) : Assumption 1 is not valid, because the statement for effective school education, parents have greater role than the government is not appropriate according to the passage. The passage does not discusses the roles of parents and government. The passage focuses on the educational challenges and expectations faced by the children.

• **Assumption 2 is also not valid,** because the passage generally points out the lags in basic academic competencies. The passage do not suggest the uniform curriculum across the country would address the issue. So this assumption is not valid according to the passage.

12. Which one of the following statements best reflects the central idea conveyed by the passage?

(a) Total eradication of poverty in the country will resolve the issue of under-performance of our school children.
(b) Monetary incentives to parents and teachers is a strategy to improve the children's academic performance.
(c) Public policy should ensure that competencies and achievements to young people are aligned with their expectations.
(d) India is not going to take advantage of the demographic dividend unless some school pass-outs go back to agriculture.

Ans. (c) : Options (A) is not correct, because the passage discusses about the educational challenges and expectation faced by the children and their families but the options 'a' discusses the total eradication of poverty in the country will resolve the issue of underperformance of school-children. So, the option is not according the scope of the passage.

• **Options (B) is not correct,** because the passage doesn't discusses the monetary incentives to parents and teacher is a strategy to improve the children is academic performance, The passage discusses the got between competencies and actual abilities. So, the options 'B' is not according to the scope of the passage.

• **Options (c) is correct,** because in the assumption 'public policies should ensure that competencies and achievement of young people are aligned with their expectations. So, the options 'c' is appropriate according to the passage.

• **Options (D) is not correct,** because the statement' India is not going to take advantage of demographic dividend unless some school pass outs go back to agriculture,' is not suitable according to the passage.

The options 'D' is against the message conveyed by the passage because in present time hardly among wants to go back for agriculture.

Passage–2

We take it for granted now that sciene has a social responsibility. The idea would not have occurred to Newton or Galileo. They thought of science as an account of the world as it is, and the only responsibility that they acknowledged was to tell the truth. The idea that science is a social enterprise is modern, and it begins at the industrial revolution. We are surprised that we cannot trace a social sense further back, because we nurse the illusion that the industrial revolution ended a golden age.

13. Which one of the following statements best reflects the thinking of the author about the science?

(a) Science must value the commitment of the scientists.
(b) Science is a product of civilized society and must be used for the promotion of scientific awareness in people.
(c) Industrial revolution was made possible by the advancements in science.
(d) Science must pursue truth but be responsible for social welfare.

Ans. (d) : Options 'a' is not correct, because the statement 'Science must value the commitment of the scientist' is not logical according to the passage. The passage focuses on the importance of scientist's commitment but doesn't address on social responsibilities of science.

• **Options 'b' is not correct,** because the statement may slightly touch the social role of science, that does not specially address the social responsibility or consequence that points out scientific awareness in people. So, the option 'B' is not appropriate according to the passage.

• **Options 'c' is not correct,** because the statement focuses the scientific advancement and industrial revolution is part of science but technically the passage doesn't focus on the casual relationship but rather on how the concept of science's social responsibilities develop around the industrial revolution.

• **Options 'd' is correct,** because the statement reflects the dual aspect of science that pursuing truth, as in the times of Newton and Galileo and also being responsible for social welfare, a moderns perspective that develop around the time of the industrial revolution. The passage focuses that the scenario of the science should be change and bring social welfare in industrial revolution. Above all statements are correct.

14. Consider the sequence
A_BCD_BBCDABC_DABC_D
that follows a certain pattern. Which one of the following completes the sequence?

(a) B, A, D, C (b) B, A, C, D
(c) A, A, C, D (d) A, A, C, D

Ans. (c) : ⇒ A _ BCD _ BBCDABC_ DABC_D
Complete pattern would be–
AABCD ABBCD ABCCD ABCDD
There option (c) A, A, C, D is correct.

15. The persons P and Q enter into a business, P puts ₹14,000 more than Q, but P has invested for 8 months and Q has invested for 10 months. If P's share is ₹ 400 more than Q's share out of the total profit of ₹ 2,000, what is the capital contributed by P?

(a) ₹ 30,000 (b) ₹ 26,000
(c) ₹ 24,000 (d) ₹ 20,000

Ans. (a) : Person P : Person Q
investment → x + 14000 : x
time (month) → 8 : 10
Profit → 1200 : 800

$$\text{So, } \frac{8(x+14000)}{10x} = \frac{1200}{800} = \frac{3}{2}$$

$$\Rightarrow \frac{4(x+14000)}{5x} = \frac{3}{2}$$

$$\Rightarrow 8x + 12000 = 15x$$

$$x = \frac{112000}{7} = 16000$$

investment by P = x + 14000 = 16000 + 14000
P = 30,000₹

16. P's salary is 20% lower than Q's salary which is 20% lower than R's salary. By how much percent is R's salary more than P' salary?

(a) 48.75% (b) 56.25%
(c) 60.50% (d) 62.25%

Ans. (b) : Let's assume salary of R, is = 100
Then salary of Q will be = 80 {20% less than R}
and salary of P will be = 64 {20% less than Q}
R's salary mark than P's is = 100 – 64 = 36

$$\text{in \% of R's salary than P's salary} = \frac{36}{64} \times 100 = 56.25\%$$

17. A number is mistakenly divided by 4 instead of multiplying by 4. What is the percentage change in the result due to this mistake?

(a) 25% (b) 50%
(c) 62.75% (d) 93.75%

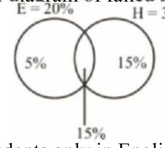
Ans. (d) : Let's assume a number is x.
Multiply 4 is = 4x
Mistakenly divided by 4 is = $\frac{x}{4}$
% change in this mistake will be

$$= \frac{\left(4x - \frac{x}{4}\right)}{4x} \times 100 = \frac{15}{16} \times 100 = 93.75\%$$

18. In an examination, 80% of students passed in English, 70% of students passed in Hindi and 15% failed in both the subjects. What is the percentage of students who failed in only one subject?

(a) 15% (b) 20%
(c) 25% (d) 35%

Ans. (b) : Students failed in English = 100 – 80 = 20%
Students failed in Hindi = 100 – 70 = 30%
Failed in both = 15%
Now Ven-diagram of failed Students –



Failed students only in English = 5%
Failed students only in Hindi = 15% Total failed student in only one subjects = 5 + 15 = 20%

19. A father said to his son, "n years back I was as old as you are now. My present age is four times your age n years back." If the sum of the present ages of the father and the son is 130 years, what is the difference of their ages?

(a) 30 years (b) 32 years
(c) 34 years (d) 36 years

Ans. (a) : Father's age Son's age
Present age → F 130 – F
n years back → F – n 130 – F– n
According to question –
F – n = 130 – F
2F = 130 + n
n = 2F – 130 .....(i)
According to question next condition–
F = 4 (130 – F – n)
F = 4(130 – F – 2F + 130)
F = 4(260 – 3F)
F = 1040 – 12F
13 F = 1040

$$F = \frac{1040}{13} = 80$$

age of son = 130 – 80 = 50
different of their ages = 80 – 50 = 30 years.

20. Consider the following:

1. 1000 litres = 1 m^3
2. 1 metric ton = 1000 kg
3. 1 hectare = 1000 m^2
Which of the above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : As we know that –
1. 1000 litres = 1 m^3
2. 1 metric ton = 1000kg
3. 1 hectare = 10,000 m^2 = 100 acres

Directions for the following 4 (four) items:
Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage–1

Today, if we consider cities such as New York, London and Paris as some of the most iconic cities in the world, it is because plans carrying a heavy systems approach were imposed on their precints. The backbone of the systems theory is the process of translating social, spatial and cultural desirables into mathematical models using computing, statistics, optimization and an algorithmic way of formulating and solving problems. The early universities of the West which began to train professionals in planning, spawned some of the most ingenious planners, who were experts in these domains. This was because these very subjects were absorbed into the planning curriculum that had its roots in the social sciences, geography and architecture. Planning in India, and its education differ from the West.

21. Which one of the following statements best reflects the most logical and rational inference that can be made from the above passage?

(a) Curriculum for urban planning courses should have diverse and interdisciplinary approach.
(b) In India, city administration is under bureacracy which lacks formal training in urban planning and management.
(c) India, the management of urban areas is a local affair with a chronic problem of insufficient funds.
(d) With high density of population and widespread poverty in our urban areas, planned development in them is very difficult.

Ans.(a): Inference (a) is correct, because in the statement "Curriculum for urban planning courses should have diverse and interdisciplinary approach, " we find that the passage points out the importance of a systems approach to urban planning, that factors in social, station and cultural aspects.

• **Inference (b) is not correct-** Because the passage does not discusses that, In India, city administration is under bureaucracy which lacks formal training in urban planning and management. So, we find that inference (b) is not logical according to the passage.

• **Inference (c) is not correct,** because the statement, "In India, the management of urban areas is a local affair with a chronic problem of insufficient funds in is illogical and beyond the scope of the passage.